FPT UNIVERSITY

LAB REPORT

ELECTRONIC CIRCUITS

Lab 1:
VOLTAGE DOUBLER

Team: <team 6>
Class: IC2004-ELC201
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HO CHI MINH CITY, JUNE 2025

I. INTRODUCTION

A full-wave voltage doubler is an essential circuit in power electronics, designed to convert an alternating current (AC) input into a direct current (DC) output with approximately twice the peak value of the input voltage. The circuit typically consists of two diodes and two capacitors arranged to utilize both the positive and negative half-cycles of the AC waveform. During operation, each capacitor charges during alternate half-cycles, and the output voltage is taken across both capacitors connected in series, effectively doubling the peak input voltage. This configuration eliminates the need for a transformer with a high turns ratio, offering a more compact and cost-efficient design. Full-wave voltage doublers are commonly employed in applications that require high-voltage, low-current DC sources, such as cathode-ray tubes, microwave oven circuits, RF communication equipment, and energy harvesting systems (Sedra & Smith, 2020). Their ability to generate elevated voltages & simultaneously converts the AC signal to DC makes them valuable in both industrial and consumer electronic systems, particularly where space and efficiency are critical considerations.

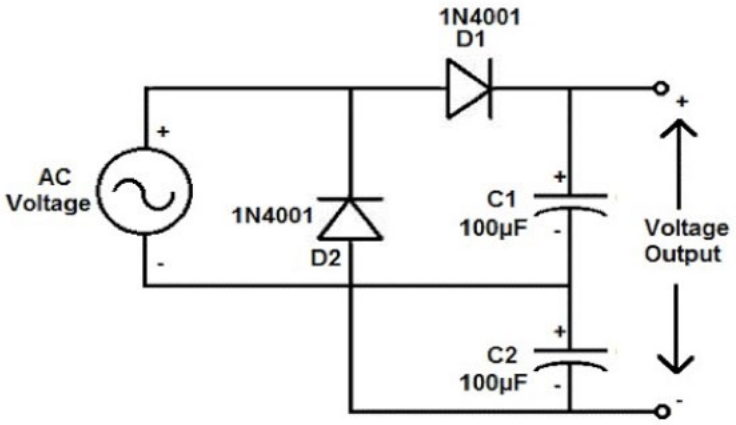


Figure 1. The circuit diagram of <*voltage doubler*>.

II. SIMULATION METHOD

1. Schematic diagram

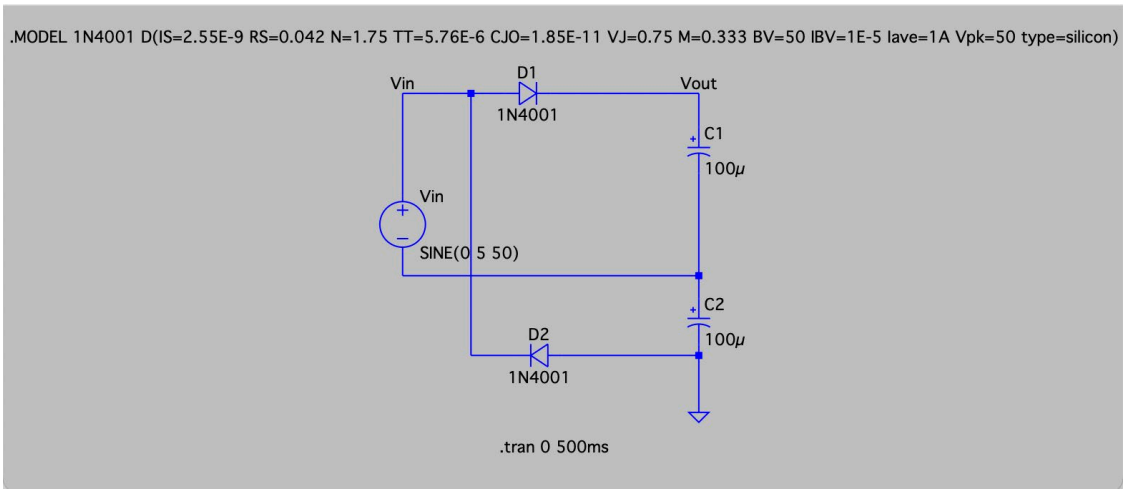


Figure 2. The schematic of <*voltage doubler*> designed using <*LTSpice*>.

1.1. Circuit Configuration

The typical full-wave voltage doubler (Delon circuit) consists of two diodes (D1, D2) and two capacitors (C1, C2) arranged with an AC input source. The output is taken across the series combination of C1 and C2. A simple circuit diagram would be essential for illustration above.

1.2. Operation During Positive Half-Cycle of AC Input (e.g., Vin>0)

Assuming a standard orientation of components: Diode D1 becomes forward-biased, and Diode D2 is reverse-biased. Current flows through D1 to charge capacitor C1 up to the peak of the input AC voltage (V_{peak}), so $V_{C1} \approx V_{peak}$ (less the forward voltage drop V_{f_D1} across D1). The polarity of C1 will be positive at the anode of D1. Capacitor C2 is not significantly charged during this half-cycle as D2 is off.

1.3. Operation During Negative Half-Cycle of AC Input (e.g., Vin<0)

During this half-cycle, Diode D2 becomes forward-biased, and Diode D1 is reverse-biased. Current flows through D2 to charge capacitor C2 up to the peak of the input AC voltage ($|V_{peak}|$), so $V_{C2} \approx |V_{peak}|$ (less the forward voltage drop V_{f_D2} across D2). The polarity of C2 will be such that the terminal connected to the cathode of D2 is positive. Capacitor C1 retains its charge from the previous half-cycle as D1 is off.

1.4. Summation of Output Voltage

The output voltage (V_{out}) is taken across the series combination of C1 and C2. Now both the capacitors C1 and C2 are charged with its maximum value V_{peak}. When load is connected in the output terminal, the capacitor will discharge in the load. Hence ideal equation will be written as:

Ideally, $$\boldsymbol{V_{out_ideal} = V_{peak} + V_{peak} = 2 \times V_{peak}.}$$

Practically, $$\boldsymbol{V_{out} \approx (V_{peak} - V_{f_D1}) + (V_{peak} - V_{f_D2}) = 2 \times V_{peak} - (V_{f_D1} + V_{f_D2}).}$$

2. Simulation settings

The simulation of the full-wave voltage doubler was carried out using **LTspice XVII**, a SPICE-based circuit simulator widely used in academia and industry due to its high accuracy and flexible model customization.

2.1 Simulation Platform Configuration

LTspice provides a user-friendly graphical interface for circuit construction and waveform analysis. In this experiment, the following setup was applied:

- **Simulation software**: LTspice XVII
- **Analysis type**: Transient analysis
- **Time range**: 0 to 500 milliseconds
- **Time step**: Automatically adjusted by LTspice

- **Command line used**: .tran 0 500ms

2.2 Step-by-Step Simulation Procedure

The following steps were taken to construct and simulate the circuit:

1. **Launch LTspice** and create a new schematic workspace.
2. **Add the components**:
   - **AC Voltage Source (V):**
     - Add a voltage source component.
     - Right-click to edit: Select "SINE" function.
     - Set DC offset = 0V.
     - Set Amplitude = 5V (this is Vpeak for LTspice's SINE source).
     - Set Freq = 50Hz.
   - **Diodes (D):**
     - Add two diode components.
     - Right-click each diode: Select "Pick New Diode."
     - Choose "1N4001" from the library. If not directly available, a generic diode model can be used, but specifying 1N4001 (if in the LTspice library) or using its model parameters will yield more accurate results.
     - **Diode model customization**: To improve the accuracy of diode behavior, a specific model for the 1N4001 diode was declared:

       MODEL 1N4001 D (IS=2.55E-9 RS=0.042 N=1.75 TT=5.76E-6 CJO=1.85E-11 VJ=0.75 M=0.333 BV=50 IBV=1E-5 Iave=1A Vpk=50; type=silicon).

     - This model incorporates key diode characteristics, including reverse recovery time, forward voltage, breakdown voltage, and junction capacitance, which ensures the transient response matches real-world behavior more closely.
   - **Capacitors (C):**
     - Add two capacitor components.
     - Right-click each capacitor: Set capacitance value to 100uF.
   - **Ground (GND):**
     - Add a ground symbol and connect it to the appropriate point in the circuit. A common representation of the Delon circuit has the common connection of C1 and C2 as the negative output, and this point is often grounded. The positive output is then at the cathode of D2 (or the common junction of D1 cathode and D2 anode, depending on drawing).

3. **Wire Components:** Connect the components according to the full-wave voltage doubler (Delon circuit) topology. Ensure correct diode orientation.
   - AC source connected between the common input point and ground.

- Common input point to D1 anode and D2 cathode.
- D1 cathode to C1 positive.
- D2 anode to C2 negative.
- C1 negative and C2 positive are connected together (midpoint).
- Positive output is C1 positive.
- Negative output is C2 negative.
- Output voltage V_{out} is measured between the positive and negative output terminals.

4. **Simulation command**:

- From the "Simulate" menu, select "Edit Simulation Cmd."
- Choose the "Transient" tab.
- Set "Stop time": To observe multiple cycles and settling, the input is 50Hz (20ms period), a stop time of 200ms to 500ms is adequate. [15] For 50Hz, 10-20 cycles (200ms-400ms) will show settling.
- "Time to start saving data": 0.
- Transient analysis command : .tran 0 500ms.

5. **Run the simulation**:
    - Press the **"Run"** button to begin the simulation.
    - Use the probe tool to click on the input node (V_{in}) and output node (V_{out}) to visualize the voltage waveform.

III. IMPLEMENTATION METHOD

1. PCB layout

(*Provide the snapshot of the designed PCB, then describe in detail the operations of the circuit from its components*)

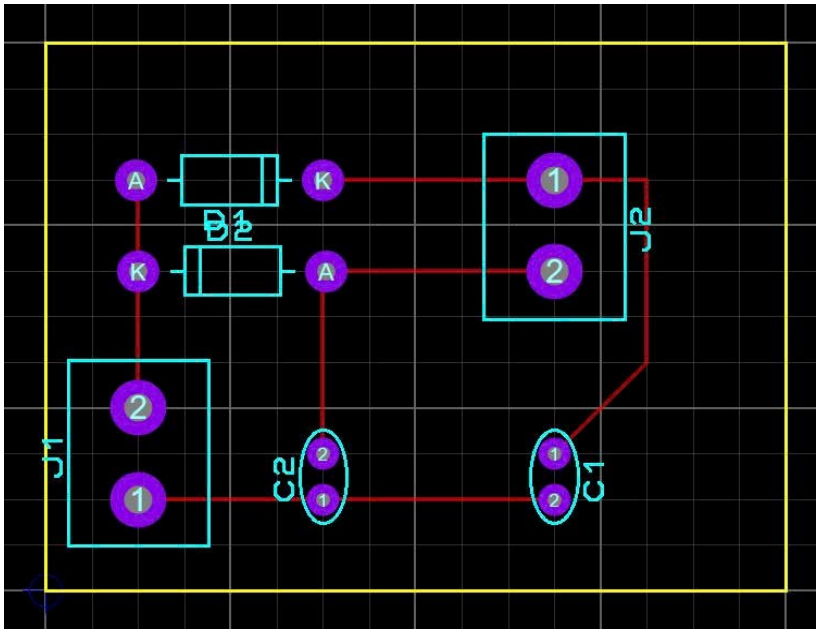


Figure 3. PCB edge size 30 x 40 mm

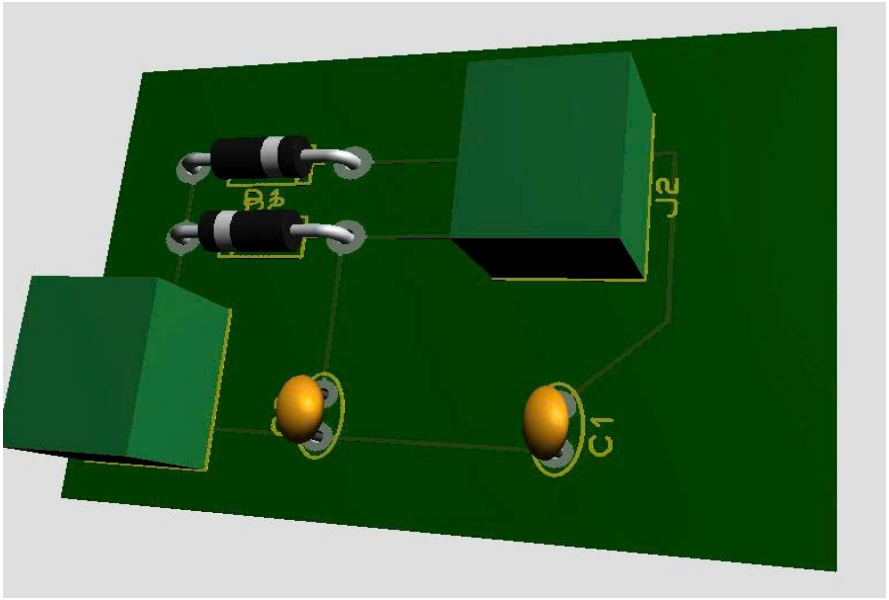


Figure 4. 3D view

The core mechanism of the full-wave voltage doubler is the independent charging of two capacitors C1,C2 during alternate half-cycles of the input, followed by their series connection to the output J2. This means each capacitor "captures" a peak voltage from one half-cycle, from the source J1, and the circuit arrangement ensures these captured voltages are then presented in output terminal J2, summing their values.

2. Final product

(*Provide the snapshot of the implemented circuit, then describe in detail its structure*)

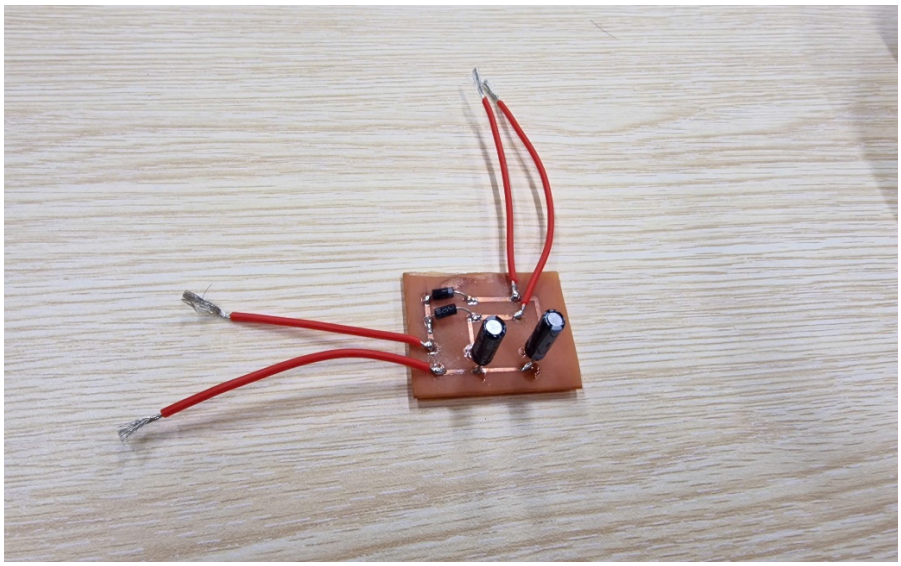


Figure 5. Assembed board of <*voltage doubler*>.

This circuit is built on a copper-clad board. The conductive paths are created using an etching process, after which the components are soldered according to the circuit diagram. These components include two 100µF capacitors (C1, C2), two 1N4001 diodes, and four wires for the input terminal J1 and output terminal J2.

IV. SIMULATION RESULTS

(*Provide simulation results to verify the operation of the circuit. Comment on these results*)

When running simulation, we could not understand why the actual result that we measured on our circuit and the simulation's are so much different from the hand-calculation's. Therefore, we put different loads across the output and observe the different behaviours of output voltage.

- **R = 1k Ohm.**

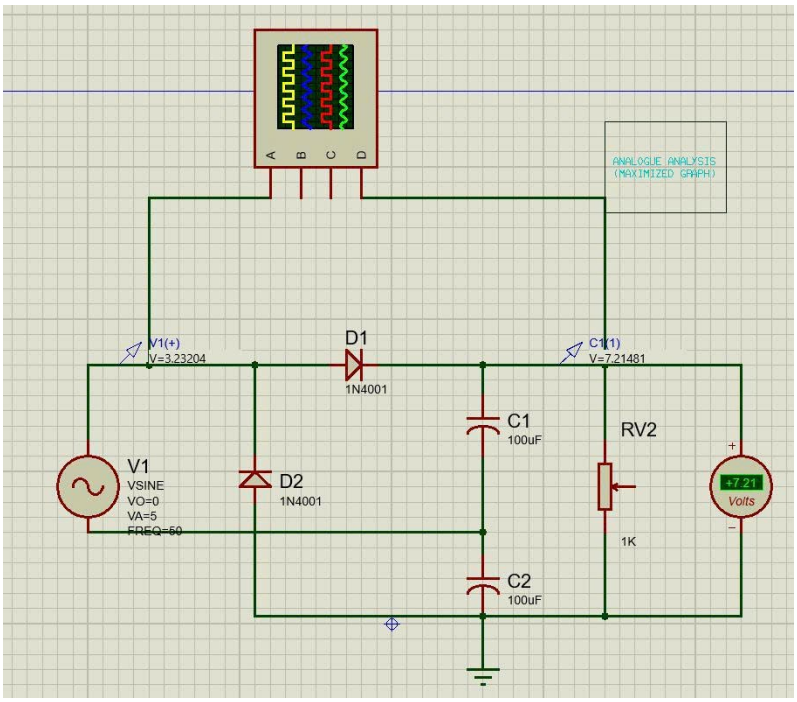


Figure 6. V_out (average) = 7.21 V

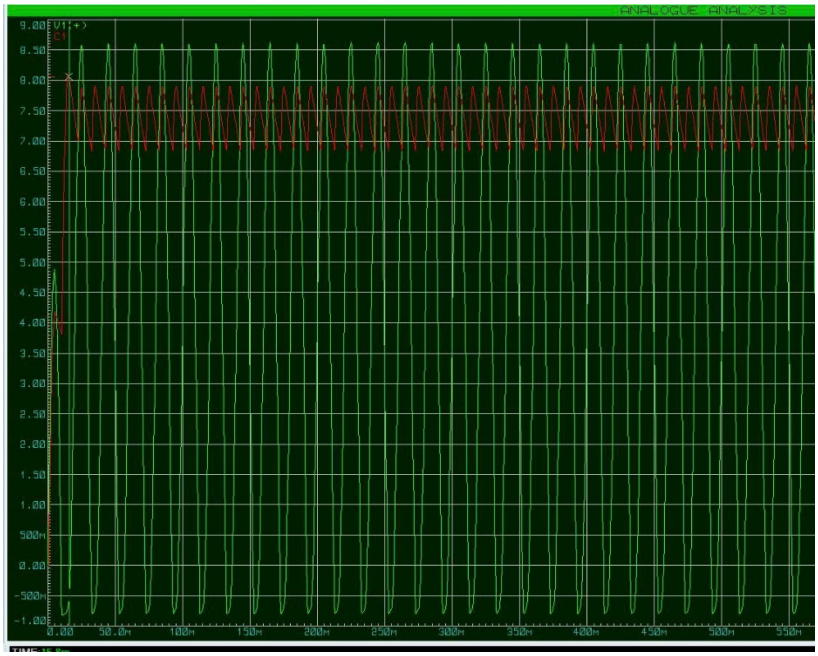


Figure 7. Oscillator reading: V_out (red line) behaves like a wave, fluctuates: 6.87 V-7.9 V

- **R = 10k Ohm.**

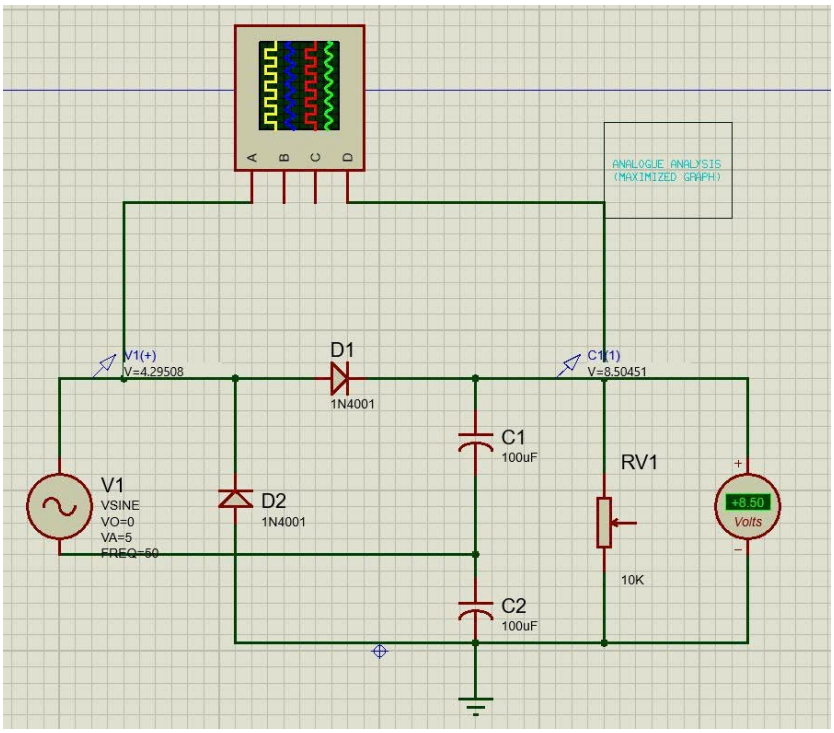


Figure 8. V_out (average) = 8.5 V

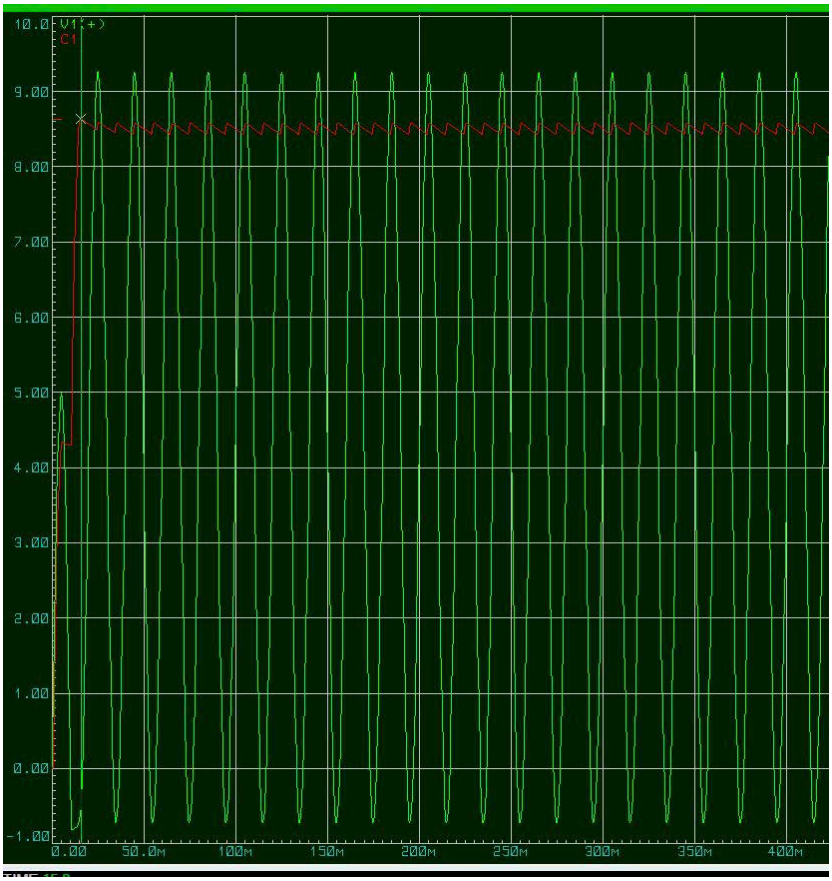


Figure 9. Oscillator reading: V_out (red line) behaves like a wave, fluctuates: 8.45 V-8.57 V

- **R = 100k Ohm**

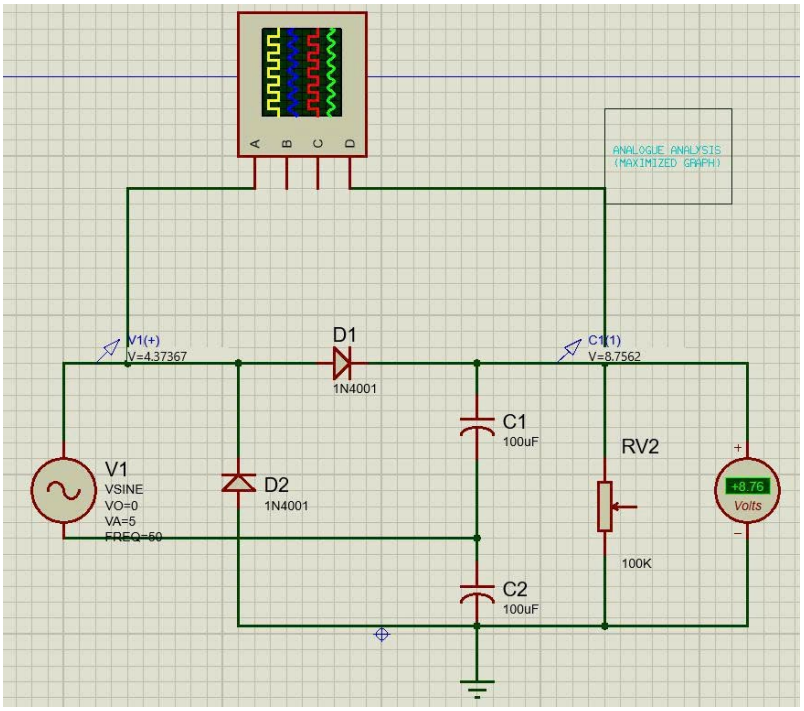


Figure 10. V_out (average) = 8.76 V

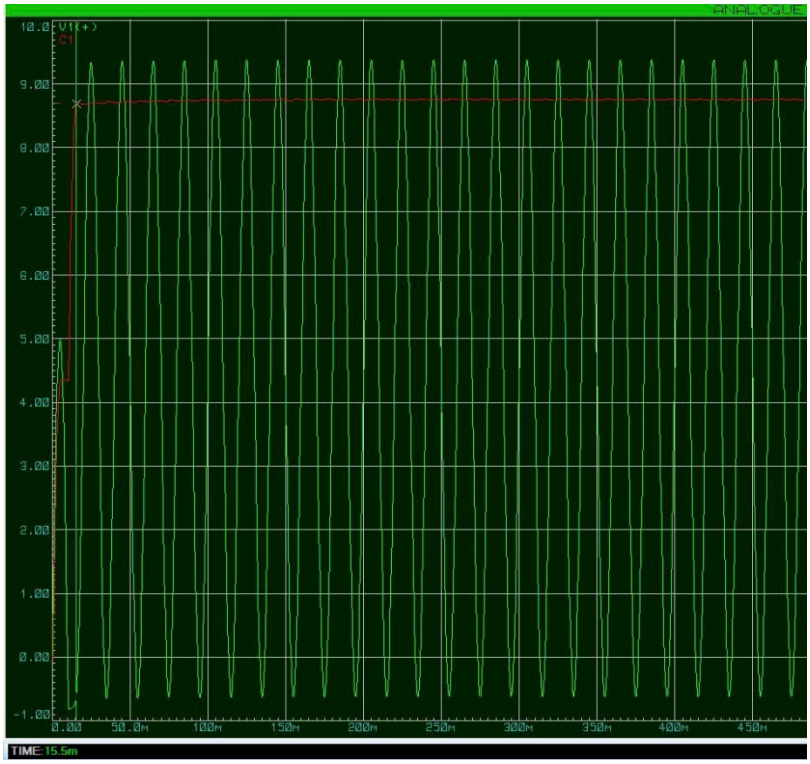


Figure 11. Oscillator reading: V_out (red line) behaves like a DC output, fluctuates between 8.75 V – 8.76 V

- **Without R (the same as our PCB)**

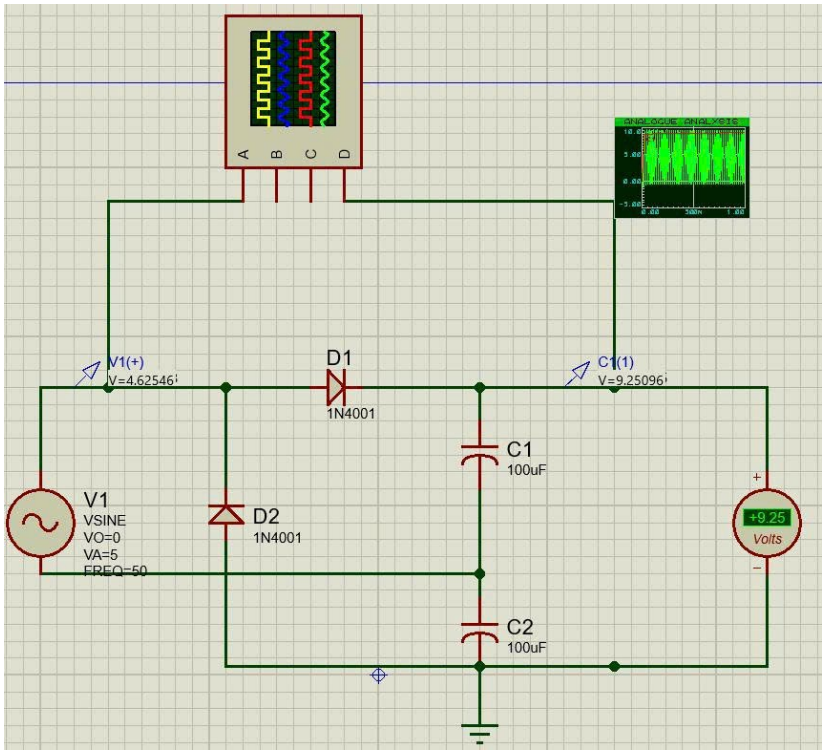


Figure 12. V_out (average) = 9.25 V

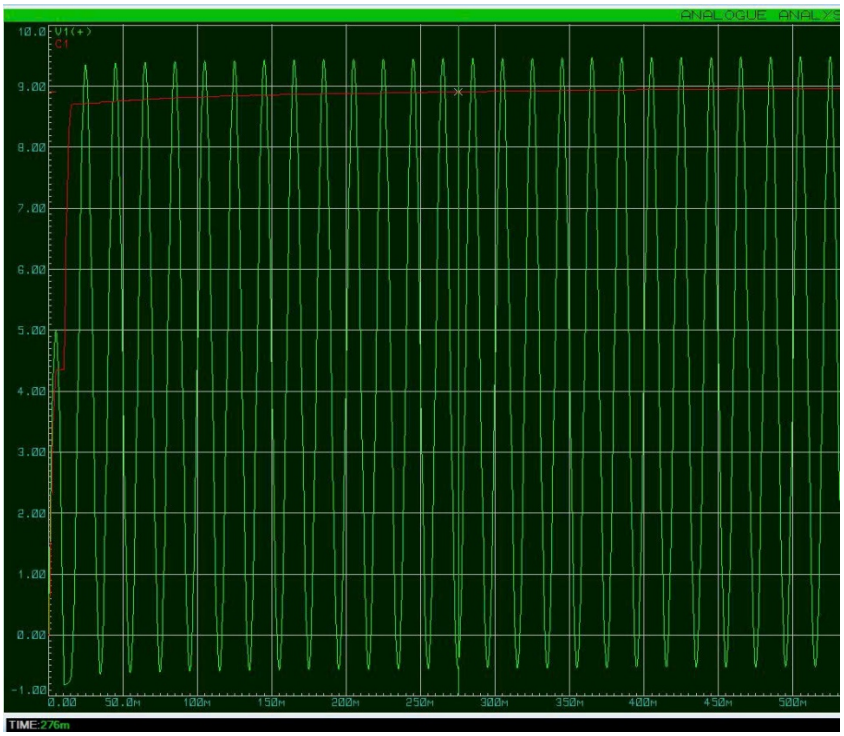


Figure 13. Oscillator reading: V_out (red line) behaves like a DC output, constant at 9.25V

Observation:

As load resistance (R_load) decreased (I_load = V_out / R_load increased), the average output voltage (V_out) dropped and ripple voltage became more pronounced. Conversely, increasing R_load led to higher V_out and reduced ripple.

Small R_load:
Capacitors discharge more between cycles.
Greater voltage drop → lower V_out (e.g., 7.21V with 1kΩ).
Larger ripple.

Large R_load:
Less discharge.
Higher V_out (e.g., 8.96V with 1MΩ).
Smaller ripple.

No Load:
Capacitors retain full charge.
Maximum V_out (~9.3V), minimal ripple.
Minor discrepancies (e.g., 9.3V vs. 9.16V) are due to simulation accuracy, diode modeling, and timing.

V. EXPERIMENTAL RESULTS

(*Provide experimental results to verify the operation of the circuit. Comment on these results and compare them with simulation results*)

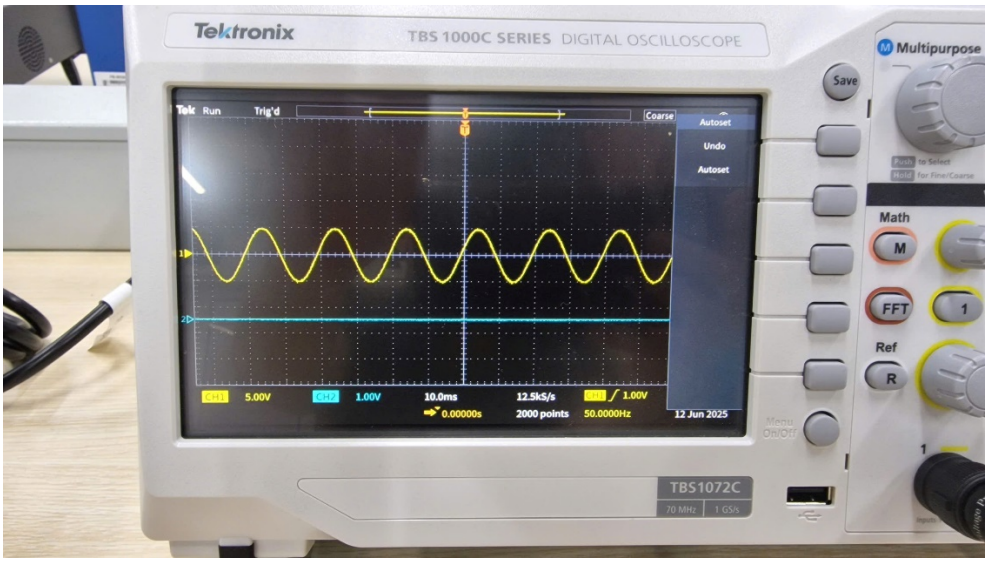


Figure 14. Input AC source: 50Hz, V_peak = 5 V

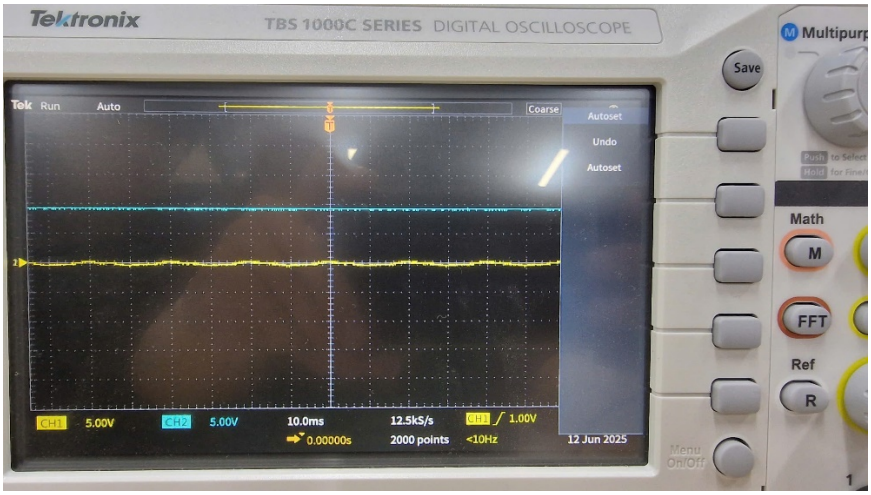


Figure 15. Output behaves as DC, observed on the oscilloscope.



Figure 16. Output measured on Voltmeter, average value

VI. CONCLUSION

1.Summary of Output Voltage and Behavior

The circuit with a 5V peak AC input, two 1N4001 diodes, and two 100uF capacitors will produce a DC output voltage of approximately 9.16V under no-load. The output will have a voltage ripple at twice the input AC frequency, although this effect is only active when a resistor is placed. The magnitude of this ripple will be relatively small with 100uF capacitors under no load, but will increase with the load current (or R_load decreases).

2. Simulation Process and Key Findings Recap

Simulation using LTspice (or similar) involves drawing the circuit schematic with accurate component models, setting up a transient analysis, and plotting the input/output waveforms. The simulation confirms the calculated DC output voltage, allows visualization of capacitor charging

and output settling, and enables measurement of the voltage ripple. It also demonstrates the effect of load on the output voltage and ripple (voltage regulation). While not varied in our test, we expect larger capacitance values would generally help reduce ripple by storing more energy and minimizing voltage drops during discharge.

3. Ripple effect:

Ripple Behavior Ripple ∝ Load Current (I_load).
Ripple ∝ 1 / Load Resistance (R_load).

- V_ripple = I_load / (f.C)
- For Fullwave Volt Doubler, each Capacitor is charged during half cycle: V_ripple = I_load / (2.f.C)

Voltage doublers are load-sensitive and best suited for light-load, high-voltage applications (e.g., biasing circuits).
Larger capacitors would reduce ripple by storing more energy.
Simulations reveal dynamic behaviors often missed in static analysis.

4. Final Remarks

The full-wave voltage doubler is an effective circuit for achieving transformerless DC voltage step-up for low-current applications. A clear understanding of component characteristics and their impact on performance, coupled with proper simulation, is key to successful design and analysis.

5. Future Directions of Circuit
Several future enhancements can be considered. Firstly, incorporating a voltage regulation stage—such as a Zener diode or low-dropout (LDO) regulator—can improve output stability under variable load conditions. Secondly, the substitution of conventional diodes with fast-recovery or Schottky diodes may improve efficiency by minimizing forward voltage drop and reducing switching losses, especially in high-frequency applications. Thirdly, integrating this voltage doubler into energy harvesting systems (e.g., for wireless sensor nodes) can leverage its ability to generate higher DC voltages from low-amplitude AC sources (Gray et al., 2009). Finally, further testing across a wider range of input frequencies, load resistances, and capacitor values would support optimization for specific application requirements, such as RF communication circuits or low-power embedded systems.

TEAM MEMBERS' CONTRIBUTION

Class: <IC2004>		**Team: <Team 6>**		**Contribution**
#	**Student ID**	**Student Name**	**Task**	**%**
1	SE203519	Vũ Quốc Anh	Simulation, PCB, research,report	20
2	SE201121	Trương Đông Hưng	Simulation, research, report	20
3	SE190907	Lâm Duy Lộc	Simulation, PCB	20
4	SE201132	Lê Nguyễn Huy Hoàng	Simulation, report	20
5	SE200937	Nguyễn Quang Huy	Simulation, research	20
Total:				**100%**

REFERENCES

[1] Sedra, A. S., & Smith, K. C. (2020). Microelectronic circuits (8th ed.). Oxford University Press.

[2] Gray, P. R., Hurst, P. J., Lewis, S. H., & Meyer, R. G. (2009). *Analysis and Design of Analog Integrated Circuits* (5th ed.). Wiley.